Thursday Aug-27-2026

Flowserve Facility:

Flowserve Cookeville Operations
1978 Foreman Drive
Cookeville, TN 38501

cmtrs@flowserve.com

To:

Valve Summary

SERIAL #	HEAT #	PO #	Description	Valve Type	Class	End Connection
B98645	G37187/08A0	1185801	4" N2345	PLG	900	RF X RF
B98645	6635	1185801	4" N2345	PLG	900	RF X RF
B90388	G37187/08A0	1185801	4" N2345	PLG	900	RF X RF
B90388	6068	1185801	4" N2345	PLG	900	RF X RF

MTR Certificate Managed By MetalTrace, Trace Applications Inc. www.TraceApps.com toll free: 1-866-420-7007

NAKODA ENGINEERS

ISO 9001 Certified

TEST CERTIFICATE AS PER EN 10204 3.1

TC No.:	0673	Date: 13/06/2025			
Customer:	FLOWSERVE INDIA CONTROLS PRIVATE LIMITED	Purchase order ref:	10125731	Date:	30-04-2025
Material Specification & year of edition	ASTM A516-25 GR.70	Invoice/Delivery chalan ref:	NE/B/0673/25-26	Date:	13-06-2025

PART DETAILS

Item Code	Size/Description	No. of Plates	Plate no.	Heat no.	Mill supplier	Mill TC reference
N2757909	COVER MCD 4"900/1500 GR.70 FIG.2349/3049	19	08A0	G37187	JINDAL STEEL & POWER LIMITED	0000260642

CHEMICAL ANALYSIS %

Specification value	C	S	P	Mn	Si	Cu	Ni	Cr	Mo	V	Cb	Ti	CE
Minimum	0.000	0.000	0.000	0.790	0.130	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Maximum	0.300	0.025	0.025	1.500	0.450	0.430	0.430	0.340	0.130	0.040	0.030	0.040	0.430
Mill TC value	0.141	0.001	0.015	1.490	0.300	0.006	0.016	0.021	0.002	0.001	0.015	0.012	0.390
Tested value	0.222	0.017	0.019	1.078	0.334	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.410

Chemical Test done at	OMEGA INSPECTION & ANALYTICAL LABORATORY	Lab TC ref:	TC NO:OIAL/2024/11/01132 DATE:16.11.2024

MECHANICAL ANALYSIS

Specification value	Yield Strength @ 0.2% offset method	Tensile Strength	Elongation in 2"	RA	Hardness (BHN)	Impact at -50 Deg F Min: 27 Joules Single specimen ; Avg : 27 Joules			
Minimum	38 Ksi (260 Mpa)	70 Ksi (485 Mpa)	21%		-				
Maximum	-	90 Ksi (620 Mpa)	-		233 BHN	1	2	3	Average
Mill TC value	46.70 Ksi (322 Mpa)	76.28 Ksi (526 Mpa)	30%		-	98	92	94	95
Tested value	49 Ksi (336 Mpa)	80 Ksi (551 Mpa)	27%		160, 155, 157				

Mechanical Test done at	OMEGA INSPECTION & ANALYTICAL LABORATORY	Lab TC ref:	TC NO:OIAL/2024/11/01132 DATE:16.11.2024

Heat Treatment condition: Normalized at 880 - 920˚C Cycle Time: 1.4minutes per mm thickness + 10minutes

MARKING

The plates are marked as "A516 GR 70", "N2757909" (G37187/08A0).

FLOWSERVE QA M. ILANGOVAN REVIEWED ☑ ACCEPTED ☐ VERIFIED ☑ REJECTED ☐

Visual Inspection conformed to FNDE008 Rev.10.
The Material conforms to ASTM A516-25 Gr.70 , NACE MR0175-2021/ISO 15156-1:2020 & Flowserve Specification RMC25024 Rev.08.

We hereby certify that the items mentioned above have been Visually, Dimensionally Inspected, Tested and found to be in accordance with Specification requirements.

NAKODA ENGINEERS CHENNAI 600 096

Supplier Quality (RAGUL.S)

NAKODA ENGINEERS CHENNAI 600 096

Approved by Supplier Quality Incharge (RAGUL.S)

REVIEWED / ~~NOTED~~ ..... ~~WITNESSED~~ TUVINDIA PRIVATE LIMITED TUVINDIA 660 R RANJITH

Customer/ TPI

CASTLEGATE INDUSTRIAL HOLDINGS PRIVATE LIMITED

S.F.No.387/1, Devanampalayam(P.O), Pollachi(TK), Coimbatore, 642 120
Phone :7397161500 04259264166, Mail :info@castlegatellc.com

CASTLEGATE

ISO 9001 : 2015 CERTIFIED COMPANY

INSPECTION CERTIFICATE
According to EN10204 - 3.1

Customer : FLOWSERVE INDIA CONTROLS PVT LTD
CHENNAI

T.C. No : FLS/6635 - 001
T.C.Date : 16/07/2025

PO.No : **10125866 Dt. 19/05/2025**

Heat No : 6635
Pour Date : 13/06/2025

Specification : **ASTM A216 Gr.WCC (UNS J02503) - 2021**

CASTING PROCESSED BY INDUCTION MELTING FURNACE

FOUNDRY IDENTIFICATION : **CG**

CHEMICAL COMPOSITION %

Element	C	Mn	Si	P	S	Cr	Ni	Mo	Cu	V	CE	N	AL
Minimum	---	---	---	---	---	---	---	---	---	---	---	---	---
Maximum	0.230	1.200	0.600	0.025	0.020	0.500	0.500	0.200	0.300	0.030	0.430	0.015	0.080
Actual	0.186	0.871	0.421	0.023	0.008	0.095	0.097	0.013	0.011	0.004	0.361	0.004	0.036

HEAT TREATMENT : NORMALISING: Temperature raised to 920°C, soaked for 4 Hrs and then atmospheric air cooled.(Cycle No:D0159/25-26Dt.28/06/2025)

MECHANICAL PROPERTIES

	Yield Strength		Tensile Strength		%Elongation	% Reduction	Hardness	Bend Test	Impact Test			
Requirements	kgf/mm²	Mpa	kgf/mm²	Mpa	GL :50 mm	In Area	HBW					
	0.2 % Rp		-		-	-	-	-				
Minimum	28.03	275	49.44	485	22	35	---	---				
Maximum	---	---	66.77	655	---	---	235	---	1	2	3	Avg
Actual	29.91	293.44	51.18	502.04	32.72	63.85	165	---	---	---	---	---

ITEM DESCRIPTION

SL.No.	Description	Drawing No.	Part No	Quantity
1	BODY CSTG 4" CL 900	00487876 REV 21	N487876WCC	12

Heat.Sl.No :1,2,3,4,5,6,7,8,9,10,11,12
MP.No : MP35397,MP35398,MP35399,MP35400,MP35401,MP35402,MP35403,MP35404,MP35405,MP35406,MP35407,MP35408 ,

Remarks:

1.This material conforms to NACE MR0175 - 2021 / ISO 15156 - 2 - 2020, FTDC001R8 & Material Spec No: RMC 90012 Rev15 & RMP 61158 Rev04.
2.Visual inspection as per FNDE-08 Rev.10 & MSS SP-55-2025 and found accepted. No surface irregularities to Type I none and Type II - XII not exceed 'a'.
3.Certified that the Castings are free from Radioactive contaminations.
4.MPI carried out as per MSS SP 53 - 2021 & FNDE 002 Rev 10 and found accepted.
5.Chemical Analysis carried out as per ASTM A751 - 2021 And found accepted.
6.Mechanical Test carried out as per ASTM A370 - 2024a. And found accepted.

REVIEWED / ~~NOTED~~
.....~~WITNESSED~~
R RANJITH
TUVINDIA PRIVATE LIMITED
660

FLOWSERVE
QA P. KARTHIKEYAN
REVIEWED ☑ ACCEPTED ☑
VERIFIED ☑ REJECTED ☐

We Confirm that the above material was manufactured,sampled,tested & inspected in accordance with the drawing,material specification,Customer order requirement and found to meet the requirements

Prepared By
BHUVANRAJ.E

Approved By
ARUN KUMAR.N

NAKODA ENGINEERS

ISO 9001 Certified

TEST CERTIFICATE AS PER EN 10204 3.1

TC No.:	0673	Date: 13/06/2025			
Customer:	FLOWSERVE INDIA CONTROLS PRIVATE LIMITED	Purchase order ref:	10125731	Date:	30-04-2025
Material Specification & year of edition	ASTM A516-25 GR.70	Invoice/Delivery chalan ref:	NE/B/0673/25-26	Date:	13-06-2025

PART DETAILS

Item Code	Size/Description	No. of Plates	Plate no.	Heat no.	Mill supplier	Mill TC reference
N2757909	COVER MCD 4"900/1500 GR.70 FIG.2349/3049	19	08A0	G37187	JINDAL STEEL & POWER LIMITED	0000260642

CHEMICAL ANALYSIS %

Specification value	C	S	P	Mn	Si	Cu	Ni	Cr	Mo	V	Cb	Ti	CE
Minimum	0.000	0.000	0.000	0.790	0.130	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Maximum	0.300	0.025	0.025	1.500	0.450	0.430	0.430	0.340	0.130	0.040	0.030	0.040	0.430
Mill TC value	0.141	0.001	0.015	1.490	0.300	0.006	0.016	0.021	0.002	0.001	0.015	0.012	0.390
Tested value	0.222	0.017	0.019	1.078	0.334	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.410

Chemical Test done at	OMEGA INSPECTION & ANALYTICAL LABORATORY	Lab TC ref:	TC NO:OIAL/2024/11/01132 DATE:16.11.2024

MECHANICAL ANALYSIS

Specification value	Yield Strength @ 0.2% offset method	Tensile Strength	Elongation in 2"	RA	Hardness (BHN)	Impact at -50 Deg F Min: 27 Joules Single specimen ; Avg : 27 Joules			
Minimum	38 Ksi (260 Mpa)	70 Ksi (485 Mpa)	21%		-				
Maximum	-	90 Ksi (620 Mpa)	-		233 BHN	1	2	3	Average
Mill TC value	46.70 Ksi (322 Mpa)	76.28 Ksi (526 Mpa)	30%		-	98	92	94	95
Tested value	49 Ksi (336 Mpa)	80 Ksi (551 Mpa)	27%		160, 155, 157				

Mechanical Test done at	OMEGA INSPECTION & ANALYTICAL LABORATORY	Lab TC ref:	TC NO:OIAL/2024/11/01132 DATE:16.11.2024

Heat Treatment condition: Normalized at 880 - 920˚C Cycle Time: 1.4minutes per mm thickness + 10minutes

MARKING

The plates are marked as "A516 GR 70", "N2757909" (G37187/08A0).

FLOWSERVE QA M. ILANGOVAN REVIEWED ☑ ACCEPTED ☐ VERIFIED ☑ REJECTED ☐

Visual Inspection conformed to FNDE008 Rev.10.
The Material conforms to ASTM A516-25 Gr.70 , NACE MR0175-2021/ISO 15156-1:2020 & Flowserve Specification RMC25024 Rev.08.

We hereby certify that the items mentioned above have been Visually, Dimensionally Inspected, Tested and found to be in accordance with Specification requirements.

NAKODA ENGINEERS CHENNAI 600 096

Supplier Quality (RAGUL.S)

NAKODA ENGINEERS CHENNAI 600 096

Approved by Supplier Quality Incharge (RAGUL.S)

REVIEWED / ~~NOTED~~ ..... ~~WITNESSED~~ TUVINDIA PRIVATE LIMITED TUVINDIA 660 R RANJITH

Customer/ TPI

CASTLEGATE INDUSTRIAL HOLDINGS PRIVATE LIMITED

S.F.No.387/1, Devanampalayam(P.O), Pollachi(TK), Coimbatore, 642 120
Phone :7397161500 04259264166, Mail :info@castlegatellc.com

CASTLEGATE

ISO 9001 : 2015 CERTIFIED COMPANY

INSPECTION CERTIFICATE
According to EN10204 - 3.1

Customer : FLOWSERVE INDIA CONTROLS PVT LTD
CHENNAI

T.C. No : FLS/6068 - 001
T.C.Date : 11/04/2025

PO.No : **10124146 Dt. 14/03/2025**

Heat No : 6068
Pour Date : 26/03/2025

Specification : **ASTM A216 Gr.WCC (UNS J02503) - 2021**

CASTING PROCESSED BY INDUCTION MELTING FURNACE

FOUNDRY IDENTIFICATION : **CG**

CHEMICAL COMPOSITION %

Element	C	Mn	Si	P	S	Cr	Ni	Mo	Cu	V	CE	N	AL
Minimum	---	---	---	---	---	---	---	---	---	---	---	---	---
Maximum	0.230	1.200	0.600	0.025	0.020	0.500	0.500	0.200	0.300	0.030	0.430	0.015	0.080
Actual	0.202	0.845	0.410	0.023	0.010	0.093	0.035	0.008	0.009	0.003	0.367	0.010	0.058

HEAT TREATMENT : NORMALISING: Temperature raised to 920°C, soaked for 4 Hrs and then atmospheric air cooled.(Cycle No:D0356/24-25Dt.22/03/2025)

MECHANICAL PROPERTIES

	Yield Strength		Tensile Strength		%Elongation	% Reduction	Hardness	Bend Test	Impact Test			
Requirements	kgf/mm²	Mpa	kgf/mm²	Mpa	GL :50 mm	In Area	HBW					
	0.2 % Rp		-		-	-	-	-				
Minimum	28.03	275	49.44	485	22	35	---	---				
Maximum	---	---	66.77	655	---	---	235	---	1	2	3	Avg
Actual	28.67	281.29	52.13	511.42	33.76	65.2	163	---	---	---	---	---

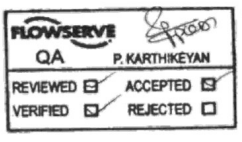
ITEM DESCRIPTION

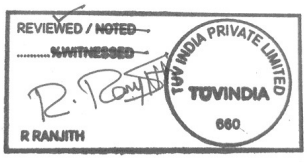
SL.No.	Description	Drawing No.	Part No	Quantity
1	BODY CSTG 4" CL 900	00487876 REV 21	N487876WCC	4

Heat.Sl.No :1,2,3,4
MP.No : MP31918,MP31919,MP31920,MP31921 ,

Remarks:

1.This material conforms to NACE MR0175 - 2021 / ISO 15156 - 2 - 2020, FTDC001R8 & Material Spec No: RMC 90012 Rev15 & RMP 61158 Rev04.
2.Visual inspection as per FNDE-08 Rev.10 & MSS SP-55-2011 and found accepted. No surface irregularities to Type I none and Type II - XII not exceed 'a'.
3.Certified that the Castings are free from Radioactive contaminations.
4.MPI carried out as per MSS SP 53 - 2021 & FNDE 002 Rev 10 and found accepted.
5.Chemical Analysis carried out as per ASTM A751 - 2021 And found accepted.
6.Mechanical Test carried out as per ASTM A370 - 2024 And found accepted.





We Confirm that the above material was manufactured,sampled,tested & inspected in accordance with the drawing,material specification,Customer order requirement and found to meet the requirements

Prepared By
BHUVANRAJ.E

Approved By
SOMASUNDARAM S

KITZ CORPORATION OF AMERICA
10750 CORPORATE DRIVE,STAFFORD,TX 77477
TEL: (281)491-7333 FAX: (281)491-9402

MESSRS.: KITZ CORPORATION OF AMERICA
DELIVER TO: HOUSTON JOB & SHEET NO.: 25J17_ 6KW19F_012_1/5
P.O. NO. #3071006329/#3071006629 DATE : OCT.29, 2025

CERTIFICATE OF COMPLIANCE

TO WHOM IT MAY CONCERN

WE HEREBY CERTIFY THAT OUR PRODUCTS AS SHOWN ON DRAWINGS OF KITZ MODEL **AK6000SCTFWZM-FS CODE 6KW19F** ARE MANUFACTURED,INSPECTED AND TESTED FULLY IN ACCORDANCE WITH THE FOLLOWING STANDARDS:

PRESSURE-TEMPERATURE RATING	1/2" TO 2"	6170 PSIG @ 300℉ MAX.
SHELL WALL THICKNESS	ANSI B16.34 CLASS 2500	
BODY & CAP	ASTM A105N	
SEAT MATERIAL	PEEK	
VALVE TESTING	API 598 ※	
FIRE TEST STANDARD	API 607 7TH EDITION	
END CONNECTION	NPT THREADS TO ANSI B1.20.1	

※ HIGH PRESSURE SHELL TEST AND LOW PRESSURE SEAT TEST ARE PERFORMED 100%

- NACE MR-0175-2002 AND MR-0103 COMPLIANCE

JY VALVE & MFG. CO., LTD.

CHARLES CHAN, PRESIDENT

KITZ CORPORATION OF AMERICA
10750 CORPORATE DRIVE,STAFFORD,TX 77477
TEL: (281)491-7333 FAX: (281)491-9402

INSPECTION CERTIFICATE

MESSRS. : KITZ CORPORATION OF AMERICA
T O : HOUSTON
P.O. NO.: : #3071006329/#3071006629

JOB & SHEET NO.: 25J17_ 6KW19F_012_2/5
EXPORT NO. : JM251003
DATE : OCT. 29, 2025

WITNESSED BY: JY VALVE & MFG. CO., LTD.

LUO, XIAODONG, QC MANAGER

VALVEMODEL	DESCRIPTION	SIZE (INCH)	Q'TY (PCS)	PRESSURE TEST PSIG			RESULT	MATERIAL	DIMENSION	VISUAL	OPERATION
				SHELL HYDRO	SEAT HYDRO	SEAT AIR					
AK6000SCTFW ZM-FS CODE 6KW19F	CLASS 2500 3-PC TYPE FORGED CARBON STEEL 316 SS TRIM THREADED END BALL VALVES	1/2"	820	9275	6800	80	GOOD	GOOD	GOOD	GOOD	GOOD

KITZ CORPORATION OF AMERICA
10750 CORPORATE DRIVE,STAFFORD,TX 77477
TEL: (281)491-7333 FAX: (281)491-9402

MATERIAL TEST RESULTS

MESSRS. : KITZ CORPORATION OF AMERICA
TO : HOUSTON
P.O. NO. : #3071006329/#3071006629

JOB & SHEET NO.: 25J17_ 6KW19F_012_3/5
EXPORT NO : JM251003
DATE : OCT. 29, 2025

VALVE MODEL : AK6000SCTFWZM-FS CODE 6KW19F SIZE : 1/2"
CLASS 2500 BALL VALVE MATERIAL: ASTM-A105N-2014
REVIEWED BY: FUJIAN LI CHUAN FORGINGS CO.,LTD.

LUO, XIAODONG, QC MANAGER
JY VALVE & MFG. CO., LTD.

CHEMICAL REQUIREMENT(%)											
ELEMENT		C	Mn	P	S	Si	Ni	Cr	Mo	Cu	V
SPEC (MAX.)		0.350	0.6-1.05	0.035	0.040	0.1-0.35	0.400	0.300	0.120	0.400	0.08
Production No	Parts Name										
25HK	BODY & CAP	0.1700	0.8800	0.0170	0.0150	0.2500	0.0700	0.0150	0.0120	0.0300	0.0170
25BK	BODY	0.1700	0.8800	0.0170	0.0150	0.2500	0.0700	0.0150	0.0120	0.0300	0.0170

TENSILE REQUIREMENTS		Tensile Strengt (MPa)	Yield Strength (MPa)	Elongation (%)	Reduction of area (%)	Hardness (HBW) Max. 187	
Production No	Parts Name	Min. 485	Min. 250	Min. 22	Min. 30	#1	#2
25HK	BODY & CAP	505	315	29	38	145	145
25BK	BODY	505	315	29	38	145	145
Heat treatment	NORMALIZING AND TEMPERING						

WE HEREBY CERTIFY THAT THE MATERIAL HEREIN DESCRIBED ARE SATISFACTORY IN ACCORDANCE WITH THE SPECIFICATION.

KITZ CORPORATION OF AMERICA
10750 CORPORATE DRIVE,STAFFORD,TX 77477
TEL: (281)491-7333 FAX: (281)491-9402

MATERIAL TEST RESULTS

MESSRS. : KITZ CORPORATION OF AMERICA
TO : HOUSTON
P.O. NO. : #3071006329/#3071006629

JOB & SHEET NO.: 25J17_ 6KW19F_012_4/5
EXPORT NO : JM251003
DATE : OCT. 29, 2025

VALVE MODEL : AK6000SCTFWZM-FS CODE 6KW19F SIZE : 1/2"
CLASS 2500 BALL VALVE MATERIAL: ASTM-A182-2018 GRADE F316
REVIEWED BY: FUZHOU SHI GANG VALVE MANUFACTURE CO.,LTD

LUO, XIAODONG, QC MANAGER
JY VALVE & MFG. CO., LTD.

CHEMICAL REQUIREMENT(%)

ELEMENT		C	Mn	P	S	Si	Ni	Cr	Mo		
SPEC (MAX.)		0.080	2.000	0.045	0.030	1.000	10.0-14.0	16.0-18.0	2.00-3.00		
Production No	Parts Name										
2505F0092	BALL	0.0280	0.9300	0.0380	0.0030	0.4100	10.0200	16.2600	2.0300		

TENSILE REQUIREMENTS		Tensile Strength (MPa)	Yield Strength (MPa)	Elongation (%)	Reduction of area (%)	Hardness (HB)
Production No	Parts Name	Min. 515	Min. 205	Min. 30	Min. 50	Max. 237
2505F0092	BALL	553	278	64	82	135
Heat treatment	SOLUTION TREATMENT					

WE HEREBY CERTIFY THAT THE MATERIAL HEREIN DESCRIBED ARE SATISFACTORY IN ACCORDANCE WITH THE SPECIFICATION.

KITZ CORPORATION OF AMERICA
10750 CORPORATE DRIVE,STAFFORD,TX 77477
TEL: (281)491-7333 FAX: (281)491-9402

MATERIAL TEST RESULTS

MESSRS. : KITZ CORPORATION OF AMERICA
TO : HOUSTON
P.O. NO. : #3071006329/#3071006629

JOB & SHEET NO.: 25J17_ 6KW19F_012_5/5
EXPORT NO : JM251003
DATE : OCT. 29, 2025

VALVE MODEL : AK6000SCTFWZM-FS CODE 6KW19F SIZE : 1/2"
CLASS 2500 BALL VALVE MATERIAL: ASTM-A276-2017 TYPE 316
REVIEWED BY: GAOYOU WEI DA STAINLESS STEEL CO., LTD.

LUO, XIAODONG, QC MANAGER
JY VALVE &MFG. CO., LTD.

CHEMICAL REQUIREMENT(%)

ELEMENT		C	Mn	P	S	Si	Cr	Ni	Mo		
SPEC (MAX.)		0.08	2.00	0.045	0.030	1.00	16.0-18.0	10.00-14.00	2.00-3.00		
Production No	Parts Name										
2505F0093	STEM	0.0510	0.6800	0.0020	0.0210	0.5500	16.4200	10.8400	2.2500		

TENSILE REQUIREMENTS		Tensile Strength (MPa)	Yield Strength (MPa)	Elongation (%)	Reduction of area (%)	Hardness (HB)
Production No	Parts Name	Min. 515	Min. 205	Min. 40	Min. 50	Max. 187
2505F0093	STEM	521	237	49	57	155
Heat treatment	COLD FINISHED					

WE HEREBY CERTIFY THAT THE MATERIAL HEREIN DESCRIBED ARE SATISFACTORY IN ACCORDANCE WITH THE SPECIFICATION.

KITZ CORPORATION OF AMERICA
10750 CORPORATE DRIVE,STAFFORD,TX 77477
TEL: (281)491-7333 FAX: (281)491-9402

MESSRS.: KITZ CORPORATION OF AMERICA
DELIVER TO: HOUSTON JOB & SHEET NO.: 25K10_ 6KW19F_012_1/5
P.O. NO. #3071006629 DATE : NOV.26, 2025

CERTIFICATE OF COMPLIANCE

TO WHOM IT MAY CONCERN

WE HEREBY CERTIFY THAT OUR PRODUCTS AS SHOWN ON DRAWINGS OF KITZ MODEL **AK6000SCTFWZM-FS CODE 6KW19F** ARE MANUFACTURED,INSPECTED AND TESTED FULLY IN ACCORDANCE WITH THE FOLLOWING STANDARDS:

PRESSURE-TEMPERATURE RATING	1/2" TO 2"	6170 PSIG @ 300℉ MAX.
SHELL WALL THICKNESS	ANSI B16.34 CLASS 2500	
BODY & CAP	ASTM A105N	
SEAT MATERIAL	PEEK	
VALVE TESTING	API 598 ※	
FIRE TEST STANDARD	API 607 7TH EDITION	
END CONNECTION	NPT THREADS TO ANSI B1.20.1	

※ HIGH PRESSURE SHELL TEST AND LOW PRESSURE SEAT TEST ARE PERFORMED 100%

- NACE MR-0175-2002 AND MR-0103 COMPLIANCE

JY VALVE & MFG. CO., LTD.

CHARLES CHAN, PRESIDENT

KITZ CORPORATION OF AMERICA
10750 CORPORATE DRIVE,STAFFORD,TX 77477
TEL: (281)491-7333 FAX: (281)491-9402

INSPECTION CERTIFICATE

MESSRS. : KITZ CORPORATION OF AMERICA
T O : HOUSTON
P.O. NO.: : #3071006629

JOB & SHEET NO.: 25K10_ 6KW19F_012_2/5
EXPORT NO. : JM251101
DATE : NOV. 26, 2025

WITNESSED BY: JY VALVE & MFG. CO., LTD.

LUO, XIAODONG, QC MANAGER

VALVEMODEL	DESCRIPTION	SIZE (INCH)	Q'TY (PCS)	PRESSURE TEST PSIG			RESULT	MATERIAL	DIMENSION	VISUAL	OPERATION
				SHELL	SEAT						
				HYDRO	HYDRO	AIR					
AK6000SCTFW ZM-FS CODE 6KW19F	CLASS 2500 3-PC TYPE FORGED CARBON STEEL 316 SS TRIM THREADED END BALL VALVES	1/2"	820	9275	6800	80	GOOD	GOOD	GOOD	GOOD	GOOD

KITZ CORPORATION OF AMERICA
10750 CORPORATE DRIVE,STAFFORD,TX 77477
TEL: (281)491-7333 FAX: (281)491-9402

MATERIAL TEST RESULTS

MESSRS. : KITZ CORPORATION OF AMERICA
TO : HOUSTON
P.O. NO. : #3071006629

JOB & SHEET NO.: 25K10_ 6KW19F_012_3/5
EXPORT NO : JM251101
DATE : NOV. 26, 2025

VALVE MODEL : AK6000SCTFWZM-FS CODE 6KW19F SIZE : 1/2"
CLASS 2500 BALL VALVE MATERIAL: ASTM-A105N-2014
REVIEWED BY: FUJIAN LI CHUAN FORGINGS CO.,LTD.

LUO, XIAODONG, QC MANAGER
JY VALVE & MFG. CO., LTD.

CHEMICAL REQUIREMENT(%)

ELEMENT		C	Mn	P	S	Si	Ni	Cr	Mo	Cu	V
SPEC (MAX.)		0.350	0.6-1.05	0.035	0.040	0.1-0.35	0.400	0.300	0.120	0.400	0.08
Production No	Parts Name										
25HK	BODY & CAP	0.1700	0.8800	0.0170	0.0150	0.2500	0.0700	0.0150	0.0120	0.0300	0.0170
25BK	BODY	0.1700	0.8800	0.0170	0.0150	0.2500	0.0700	0.0150	0.0120	0.0300	0.0170
24GK	CAP	0.1700	0.8800	0.0170	0.0150	0.2500	0.0700	0.0150	0.0120	0.0300	0.0170
25DK	CAP	0.1700	0.8800	0.0170	0.0150	0.2500	0.0700	0.0150	0.0120	0.0300	0.0170

TENSILE REQUIREMENTS		Tensile Strengt (MPa)	Yield Strength (MPa)	Elongation (%)	Reduction of area (%)	Hardness (HBW) Max. 187	
Production No	Parts Name	Min. 485	Min. 250	Min. 22	Min. 30	#1	#2
25HK	BODY & CAP	505	315	29	38	145	145
25BK	BODY	505	315	29	38	145	145
24GK	CAP	505	315	29	38	145	145
25DK	CAP	505	315	29	38	145	145
Heat treatment	NORMALIZING AND TEMPERING						

WE HEREBY CERTIFY THAT THE MATERIAL HEREIN DESCRIBED ARE SATISFACTORY IN ACCORDANCE WITH THE SPECIFICATION.

KITZ CORPORATION OF AMERICA
10750 CORPORATE DRIVE,STAFFORD,TX 77477
TEL: (281)491-7333 FAX: (281)491-9402

MATERIAL TEST RESULTS

MESSRS. : KITZ CORPORATION OF AMERICA
TO : HOUSTON
P.O. NO. : #3071006629

JOB & SHEET NO.: 25K10_ 6KW19F_012_4/5
EXPORT NO : JM251101
DATE : NOV. 26, 2025

VALVE MODEL : AK6000SCTFWZM-FS CODE 6KW19F SIZE : 1/2"
CLASS 2500 BALL VALVE MATERIAL: ASTM-A182-2018 GRADE F316
REVIEWED BY: FUZHOU SHI GANG VALVE MANUFACTURE CO.,LTD

LUO, XIAODONG, QC MANAGER
JY VALVE & MFG. CO., LTD.

CHEMICAL REQUIREMENT(%)

ELEMENT		C	Mn	P	S	Si	Ni	Cr	Mo		
SPEC (MAX.)		0.080	2.000	0.045	0.030	1.000	10.0-14.0	16.0-18.0	2.00-3.00		
Production No	Parts Name										
2505F0092	BALL	0.0280	0.9300	0.0380	0.0030	0.4100	10.0200	16.2600	2.0300		

TENSILE REQUIREMENTS		Tensile Strength (MPa)	Yield Strength (MPa)	Elongation (%)	Reduction of area (%)	Hardness (HB)
Production No	Parts Name	Min. 515	Min. 205	Min. 30	Min. 50	Max. 237
2505F0092	BALL	553	278	64	82	135
Heat treatment	SOLUTION TREATMENT					

WE HEREBY CERTIFY THAT THE MATERIAL HEREIN DESCRIBED ARE SATISFACTORY IN ACCORDANCE WITH THE SPECIFICATION.

KITZ CORPORATION OF AMERICA
10750 CORPORATE DRIVE,STAFFORD,TX 77477
TEL: (281)491-7333 FAX: (281)491-9402

MATERIAL TEST RESULTS

MESSRS. : KITZ CORPORATION OF AMERICA
TO : HOUSTON
P.O. NO. : #3071006629

JOB & SHEET NO.: 25K10_ 6KW19F_012_5/5
EXPORT NO : JM251101
DATE : NOV. 26, 2025

VALVE MODEL : AK6000SCTFWZM-FS CODE 6KW19F SIZE : 1/2"
CLASS 2500 BALL VALVE MATERIAL: ASTM-A276-2017 TYPE 316
REVIEWED BY: GAOYOU WEI DA STAINLESS STEEL CO., LTD.

LUO, XIAODONG, QC MANAGER
JY VALVE &MFG. CO., LTD.

CHEMICAL REQUIREMENT(%)											
ELEMENT		C	Mn	P	S	Si	Cr	Ni	Mo		
SPEC (MAX.)		0.08	2.00	0.045	0.030	1.00	16.0-18.0	10.00-14.00	2.00-3.00		
Production No	Parts Name										
2505F0093	STEM	0.0510	0.6800	0.0020	0.0210	0.5500	16.4200	10.8400	2.2500		

TENSILE REQUIREMENTS		Tensile Strength (MPa)	Yield Strength (MPa)	Elongation (%)	Reduction of area (%)	Hardness (HB)
Production No	Parts Name	Min. 515	Min. 205	Min. 40	Min. 50	Max. 187
2505F0093	STEM	521	237	49	57	155
Heat treatment	COLD FINISHED					

WE HEREBY CERTIFY THAT THE MATERIAL HEREIN DESCRIBED ARE SATISFACTORY IN ACCORDANCE WITH THE SPECIFICATION.

Passed Pressure Test Report

Gulf Coast Modification, L.P.
22806 Northwest Lake Drive
Houston TX 77095
USA

01 2176987

Date: 09/14/2026 Work Order#: 209127 Line #: 01

Customer PO#: 1177545

customer address

work performed

NORDSTROM N2345 W/ LOCKING DEVICE

Modify from Flanged end to Buttweld end: (0.337 W.T)
Perform RT inspection of Buttweld ends
Per: ASME B16.34 APPENDIX
INSTALL GCM SUPPLIED GEAR OP w/ HW & LOCKING DEVICE

type of test: API 6D

tested by: Ever Zavala

line item description: 4" 900 NORDSTRM WCC Plug

item#	*body heat#*	*bonnet heat#*	*serial#*	*notes*
2HA947	MP3921		61-B90388	

The above items were tested in accordance with and passed all the criterion as listed below API 6D

Test parameters: **L/P Seats:** N/A Duration: N/A **Backseat:** N/A Duration: N/A
H/P Seats: 2445 PSI Duration: 5 MIN **Shell:** 3400 PSI Duration: 15 MIN

Allowable leakage rate is 0 / No leak was detected during testing
Performed Rimpull at MAOP > 10 FTLBS / 68 turns from Fully Closed to Fully Open
Guage ID# GCM1/10018 EXP: 10/21/2026 / Guage ID# GCM1/6021 EXP:11/11/2026
Chart ID# 911513 EXP: 12/24/26 / Torque ID# TWD-GCM1-21 EXP:02/12/27

Test certified by: Ever Zavala

Ever Zavala

Test witnessed by: _____ of _____

Passed Pressure Test Report

Gulf Coast Modification, L.P.
22806 Northwest Lake Drive
Houston TX 77095
USA

TEL 713-896-3000
FAX 713-896-0683

Date: 09/14/2026 Work Order#: 209127 Line #: 02

Customer PO#: 1177545

customer address

work performed

NORDSTROM N2345 W/ LOCKING DEVICE

Modify from Flanged end to Buttweld end: (0.337 W.T)
Perform RT inspection of Buttweld ends
Per: ASME B16.34 APPENDIX
INSTALL GCM SUPPLIED GEAR OP w/ HW & LOCKING DEVICE

type of test: API 6D

tested by: Ever Zavala

line item description: 4" 900 NORDSTRM WCC Plug

item#	*body heat#*	*bonnet heat#*	*serial#*	*notes*
2HA948	MP3634		61-B98645	

The above items were tested in accordance with and passed all the criterion as listed below API 6D

Test parameters: **L/P Seats:** N/A Duration: N/A **Backseat:** N/A Duration: N/A
H/P Seats: 2445 PSI Duration: 5 MIN **Shell:** 3400 PSI Duration: 15 MIN

Allowable leakage rate is 0 / No leak was detected during testing
Performed Rimpull at MAOP > 8 FTLBS / 67 turns from Fully Closed to Fully Open
Guage ID# GCM1/10018 EXP: 10/21/2026 / Guage ID# GCM1/6021 EXP:11/11/2026
Chart ID# 911513 EXP: 12/24/26 / Torque ID# TWD-GCM1-21 EXP:02/12/27

Test certified by: Ever Zavala

Ever Zavala

Test witnessed by: _____ of _____

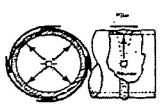
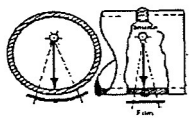
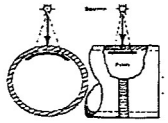
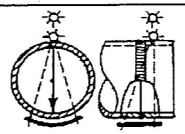
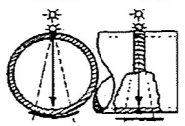
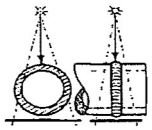
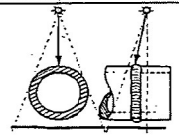
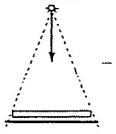
PROtect

Project Information					
Customer:			Date:	8/30/26	
Project Name:	2HA947 & 2HA948		Location:	15-801	
Unit/System:	209127		PROtect Job #:	41163	
P.O. Number:	127928		PROtect Tkt. #:	T-498935	
Customer Reference Number(s) (If Applicable):	4" 600 Nordstrom WCC Plug Valve BWE				
Inspection Procedure and Revision:	QCP 8234 REV.9	Acceptance Standard:	ASME B16.34 Appendix I LE	Report Number:	41163-CRT

Equipment Information					
Image Plate Manufacturer:	Waygate	Scanner Manufacturer:	Waygate	Scanner Serial Number:	01654
Image Plate Model:	iPS	Scanner Model:	CRxVision	Software Version/Revision:	2.4.3.26056

Technique ID	1				
Material Type	C/S				
Nominal Thickness (in inches)	0.337				
Thickness Including Reinforcement (in inches)	0.587				
Pipe / Specimen O.D.	4.500				
Exposure Geometry Sketch Letter (From Below)	B				
Exposure Type	Single Wall Exposure (SWE)				
Viewing	Single Wall Viewing (SWV)				
Radiation Source Type	Gamma				
Energy/Isotope	Ir-192				
Focal Spot Size (in inches)	0.124				
Pb Screen Thickness Front/Back:	0.010" / 0.010"				
Source to Object Distance (in inches)	4.200				
Object to Image Plate Distance (in inches)	0.587				
Ug: (in inches)	0.017				
# of Exposures for Complete Weld	4				
IQI Position	Source Side				
IQI Type	ASTM SE747 Wire Type				
IQI Material	Stainless Steel				
IQI Set/Number	ASTM B				
Sensitivity Level Required	(8) - (0.016")				
Exposure Time	00:00:06				

Exposure Geometry

SWE / SWV Techniques			DWE / SWV Techniques		DWE / DWV Techniques		Plate
							
A	B	C	D	E	F	G	H

Comments

NDT Technician: David Kincaid _____ (Signature) II (Cert Level) 8/30/2026 (Date)

Customer Approval: _____ _____ (Print) (Signature)

Customer Approval: _____ _____ (Print) (Signature)

All information contained in this report represents test results at the time of the services. This report does not constitute a representation, warranty or guarantee of the actual condition of the materials or items tested. Company assumes all risks associated with any action or inaction it chooses to undertake as a result of the interpretation of the information provided in the report. The recording of false, fictitious, or fraudulent statements or entries on this document may be punishable as a felony under Federal Statue. This report is property of PROtect and it shall not be reproduced in full, or in part, without the written approval of PROtect, LLC. Copyright by PROtect, LLC.

Project Information			
Customer:		Date:	8/30/26
Project Name:	2HA947 & 2HA948	Location:	15-801
Unit/System:	209127	PROtect Job #:	41163
P.O. Number:	127928	PROtect Tkt. #:	T-498935
Customer Reference Number(s) (If Applicable):	4" 600 Nordstrom WCC Plug Valve BWE		
RT inspection of machined Butt Weld Ends(BWE) & up to 3" over.			

NDT Technician: David Kincaid _____ (Signature) II (Cert Level) 8/30/2026 (Date)

Customer Approval: _____ (Print) _____ (Signature)

Customer Approval: _____ (Print) _____ (Signature)

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Project Information			
Customer:		Date:	8/30/26
Project Name:	2HA947 & 2HA948	Location:	15-801
Unit/System:	209127	PROtect Job #:	41163
P.O. Number:	127928	PROtect Tkt. #:	T-498935
Customer Reference Number(s) (If Applicable):	4" 600 Nordstrom WCC Plug Valve BWE		

Inspection Procedure and Revision:	QCP 8234 REV.9	Acceptance Standard:	ASME B16.34 Appendix I LE	Report Number:	41163-CRT

Examination Results

Weld / Component ID (Serial Number)	Weld / Part Size		View	Accept	Reject	Gas/Porosity	Sand/Slag	Shrinkage Type/lvl	Shrinkage Type/lvl	Shrinkage Type/lvl	Shrinkage Type/lvl	Crack/Hot Tear	Foreign Material	Surface	Insert	Other - Explain	Other - Explain	Artifact	Technique ID	Window	Level	Filters	File Name / Comments
	Dia.	Thick.																					
2HA947 Inlet	4.5"	.337"	1-2	✓															1	65536	32768	Flash!W	2HA947_Inlet@V1
2HA947 Inlet	4.5"	.337"	2-3	✓															1	65536	32768	Flash!W	2HA947_Inlet@V2
2HA947 Inlet	4.5"	.337"	3-4	✓															1	65536	32768	Flash!W	2HA947_Inlet@V3
2HA947 Inlet	4.5"	.337"	4-1	✓															1	65536	32768	Flash!W	2HA947_Inlet@V4
2HA947 Outlet	4.5"	.337"	1-2	✓															1	65536	32768	Flash!W	2HA947_Outlet@V1
2HA947 Outlet	4.5"	.337"	2-3	✓															1	65536	32768	Flash!W	2HA947_Outlet@V2
2HA947 Outlet	4.5"	.337"	3-4	✓															1	65536	32768	Flash!W	2HA947_Outlet@V3
2HA947 Outlet	4.5"	.337"	4-1	✓															1	65536	32768	Flash!W	2HA947_Outlet@V4
2HA948 Inlet	4.5"	.337"	1-2	✓															1	65536	32768	Flash!W	2HA948_Inlet@V1
2HA948 Inlet	4.5"	.337"	2-3	✓															1	65536	32768	Flash!W	2HA948_Inlet@V2
2HA948 Inlet	4.5"	.337"	3-4	✓															1	65536	32768	Flash!W	2HA948_Inlet@V3
2HA948 Inlet	4.5"	.337"	4-1	✓															1	65536	32768	Flash!W	2HA948_Inlet@V4
2HA948 Outlet	4.5"	.337"	1-2	✓															1	65536	32768	Flash!W	2HA948_Outlet@V1
2HA948 Outlet	4.5"	.337"	2-3	✓															1	65536	32768	Flash!W	2HA948_Outlet@V2
2HA948 Outlet	4.5"	.337"	3-4	✓															1	65536	32768	Flash!W	2HA948_Outlet@V3
2HA948 Outlet	4.5"	.337"	4-1	✓															1	65536	32768	Flash!W	2HA948_Outlet@V4

Comments

NDT Technician: David Kincaid _____ (Signature) II (Cert Level) 8/30/2026 (Date)

Customer Approval: _____ _____

Customer Approval: _____ (Print) _____ (Signature)

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Passed Pressure Test Report

Gulf Coast Modification, L.P.
22806 Northwest Lake Drive
Houston TX 77095
USA

TEL 713-896-3000
FAX 713-896-0683

Date: 09/14/2026 Work Order#: 209127 Line #: 01

Customer: AIV, LP Customer PO#: 1177545

customer address

22806 NORTHWEST LAKE DRIVE HOUSTON TX 77095

work performed

NORDSTROM N2345 W/ LOCKING DEVICE

Modify from Flanged end to Buttweld end: (0.337 W.T)
Perform RT inspection of Buttweld ends
Per: ASME B16.34 APPENDIX
INSTALL GCM SUPPLIED GEAR OP w/ HW & LOCKING DEVICE

type of test: API 6D

tested by: Ever Zavala

line item description

4" 900 NORDSTRM WCC Plug

item#	*body heat#*	*bonnet heat#*	*serial#*	*notes*
2HA947	MP3921		61-B90388	

The above items were tested in accordance with and passed all the criterion as listed below API 6D

Test parameters: **L/P Seats:** N/A Duration: N/A **Backseat:** N/A Duration: N/A
H/P Seats: 2445 PSI Duration: 5 MIN **Shell:** 3400 PSI Duration: 15 MIN

Allowable leakage rate is 0 / No leak was detected during testing
Performed Rimpull at MAOP > 10 FTLBS / 68 turns from Fully Closed to Fully Open
Guage ID# GCM1/10018 EXP: 10/21/2026 / Guage ID# GCM1/6021 EXP:11/11/2026
Chart ID# 911513 EXP: 12/24/26 / Torque ID# TWD-GCM1-21 EXP:02/12/27

Test certified by: Ever Zavala
Ever Zavala

Test witnessed by: _____ of _____

Roger Johnson
Roger Johnson (Sep 23, 2026 13:57:08 CDT)

Test Object Identification

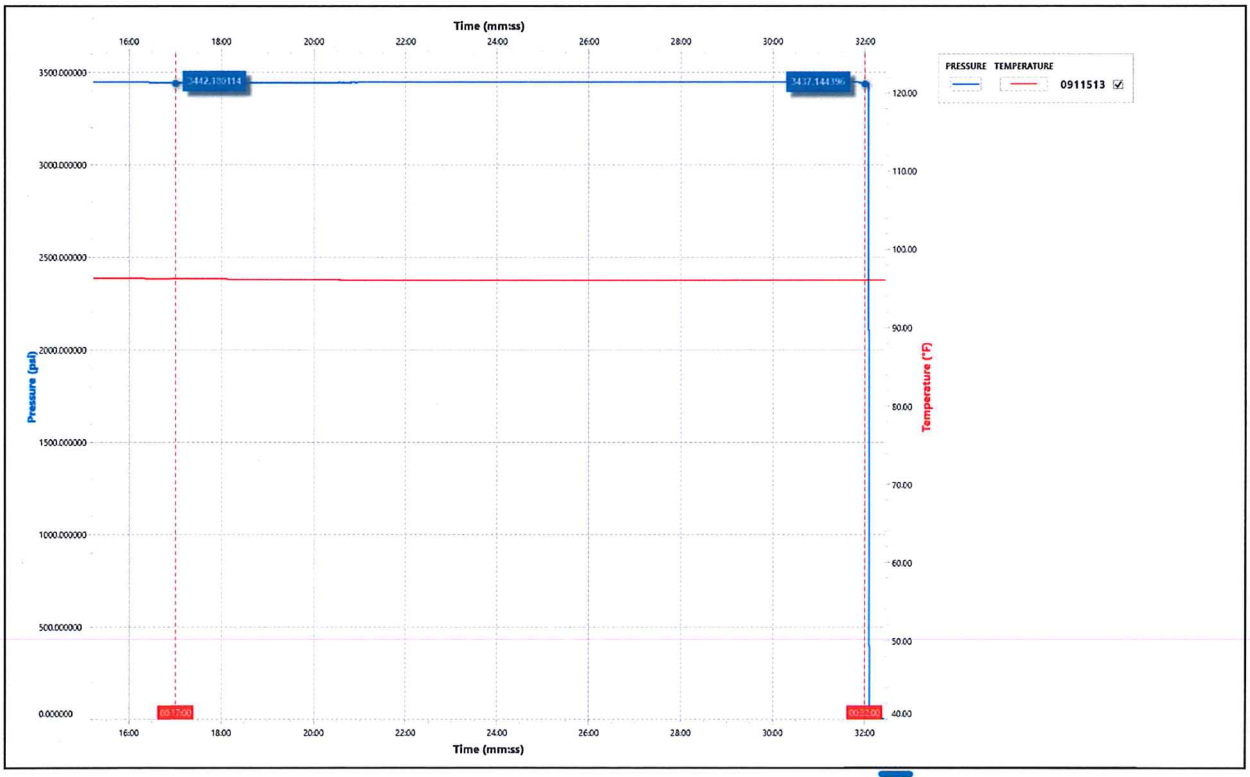
Customer:	**AIV, LP**		
Customer PO:	1177545	Serial Number:	61-B90388
Work Order:	209127-01	Valve ID:	2HA947

Certification Information

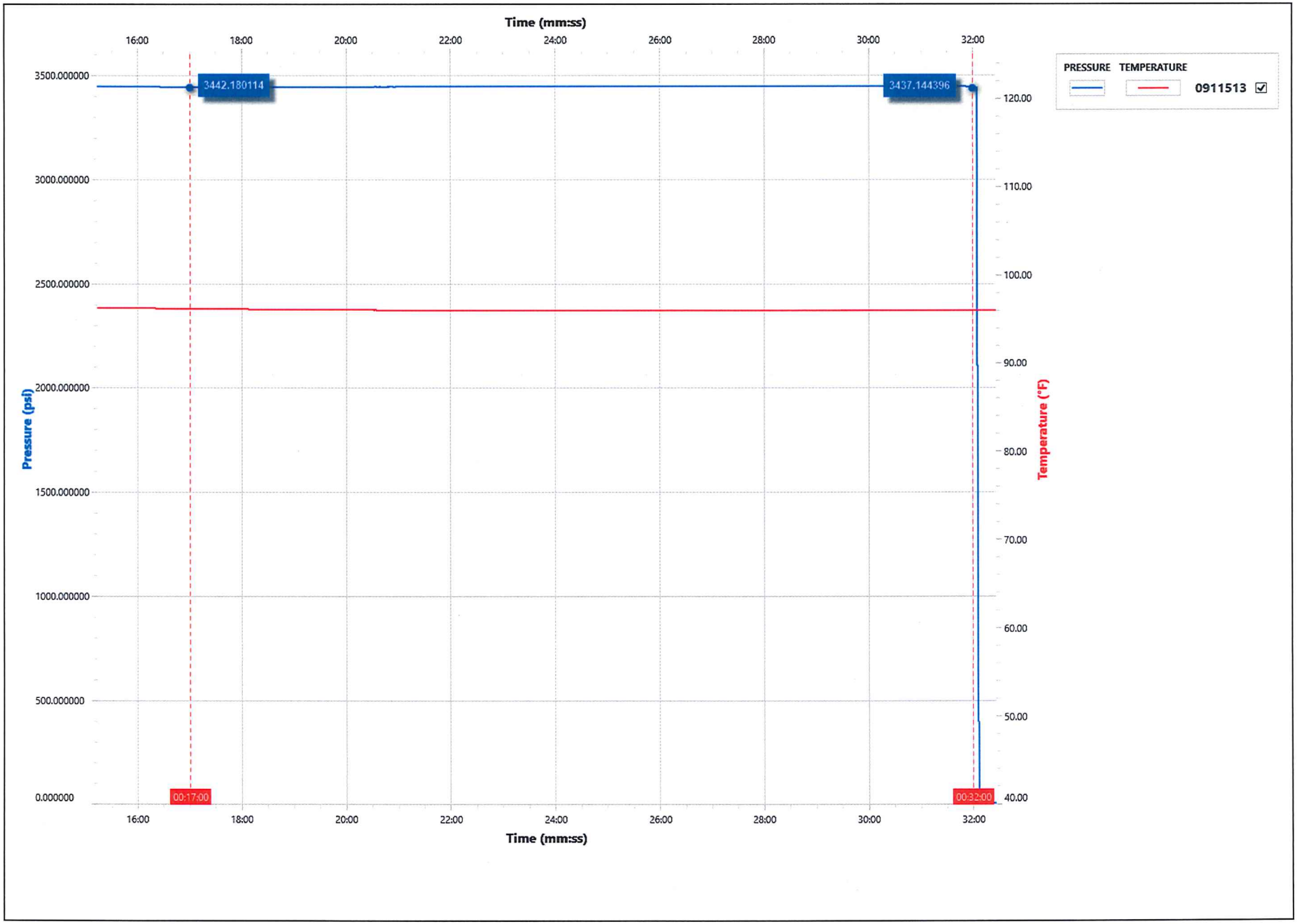
Cal. Device Used:	GCM1-0911513	Manufacturer:	ESI Technology Ltd.
Serial Number:	0911513	Calibration Date:	24/06/2026 10:45:12

Test Information

Test Date:	14/09/2026 13:55:44	Test Technician:	Ever Zavala
Procedure:	API 6D - Shell	Test Medium:	Water
Test Pressure:	3,400 PSIG	Test Duration:	15 Min



Test Technician:	Ever Zavala	Signature:	Josh Alcoser Gulf Coast Modification Josh Alcoser Inspection Coordinator
Witness/Inspector:	Roger Johnson	Signature:	Roger Johnson Roger Johnson (Sep 23, 2026 13:57:08 CDT)


Time (mm:ss)
16:00
18:00
20:00
22:00
24:00
26:00
28:00
30:00
32:00
3500.000000
3000.000000
2500.000000
2000.000000
1500.000000
1000.000000
500.000000
0.000000
Pressure (psi)
120.00
110.00
100.00
90.00
80.00
70.00
60.00
50.00
40.00
Temperature (°F)
3442.180114
3437.144396
00:17:00
00:32:00
PRESSURE
TEMPERATURE
0911513

Test Object Identification

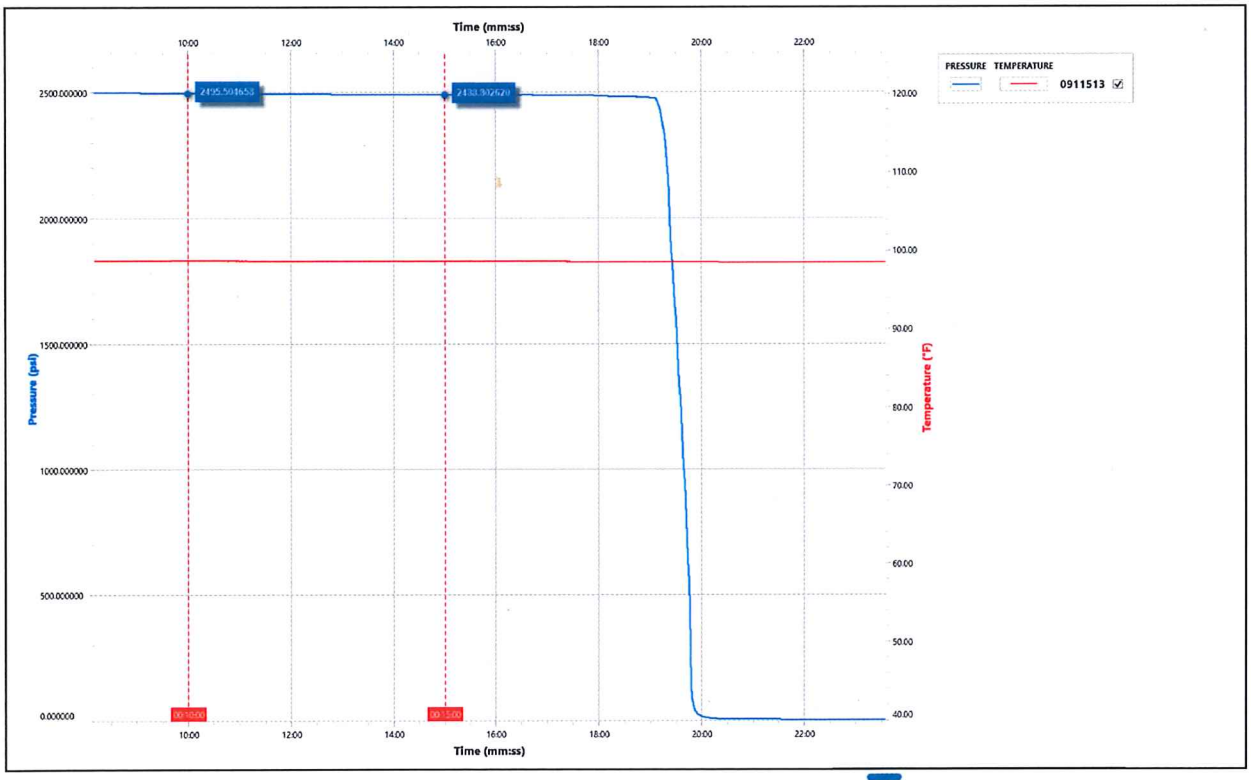
Customer:	**AIV, LP**		
Customer PO:	117545	Serial Number:	61 - B90388
Work Order:	209127-01	Valve ID:	2HA947

Certification Information

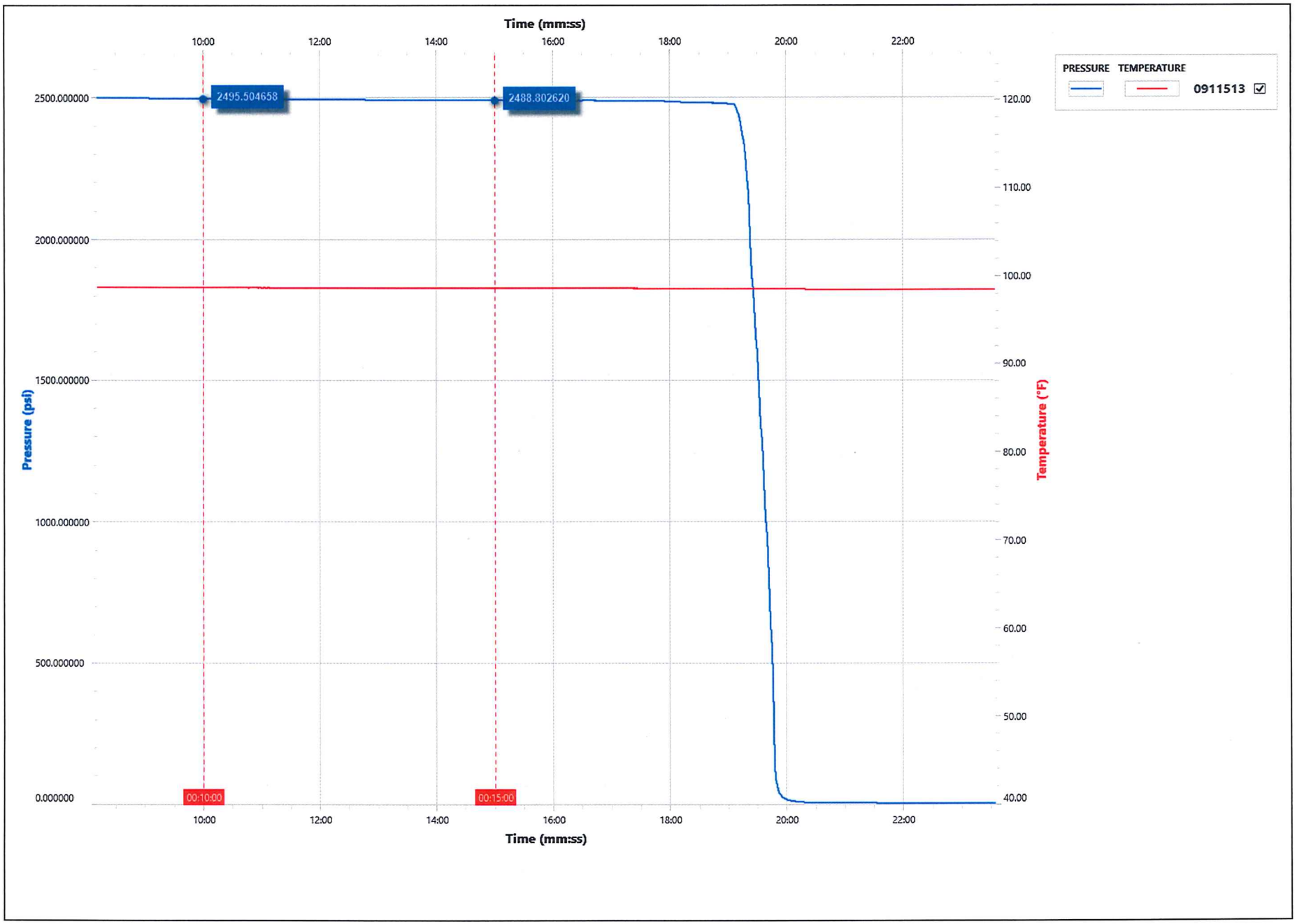
Cal. Device Used:	GCM1-0911513	Manufacturer:	ESI Technology Ltd.
Serial Number:	0911513	Calibration Date:	24/06/2026 10:45:12

Test Information

Test Date:	14/09/2026 16:07:21	Test Technician:	Ever Zavala
Procedure:	API 6D - HP Seat A	Test Medium:	Water
Test Pressure:	2,445 PSIG	Test Duration:	5 Min



Test Technician:	Ever Zavala	Signature:	Josh Alcoser Gulf Coast Modification Josh Alcoser Inspection Coordinator
Witness/Inspector:	Roger Johnson	Signature:	Roger Johnson Roger Johnson (Sep 23, 2026 13:57:08 CDT)


Time (mm:ss)
10:00
12:00
14:00
16:00
18:00
20:00
22:00
PRESSURE
TEMPERATURE
0911513
2495.504658
2488.802620
2500.000000
2000.000000
1500.000000
1000.000000
500.000000
0.000000
Pressure (psi)
120.00
110.00
100.00
90.00
80.00
70.00
60.00
50.00
40.00
Temperature (°F)
00:10:00
00:15:00
Time (mm:ss)

Test Object Identification

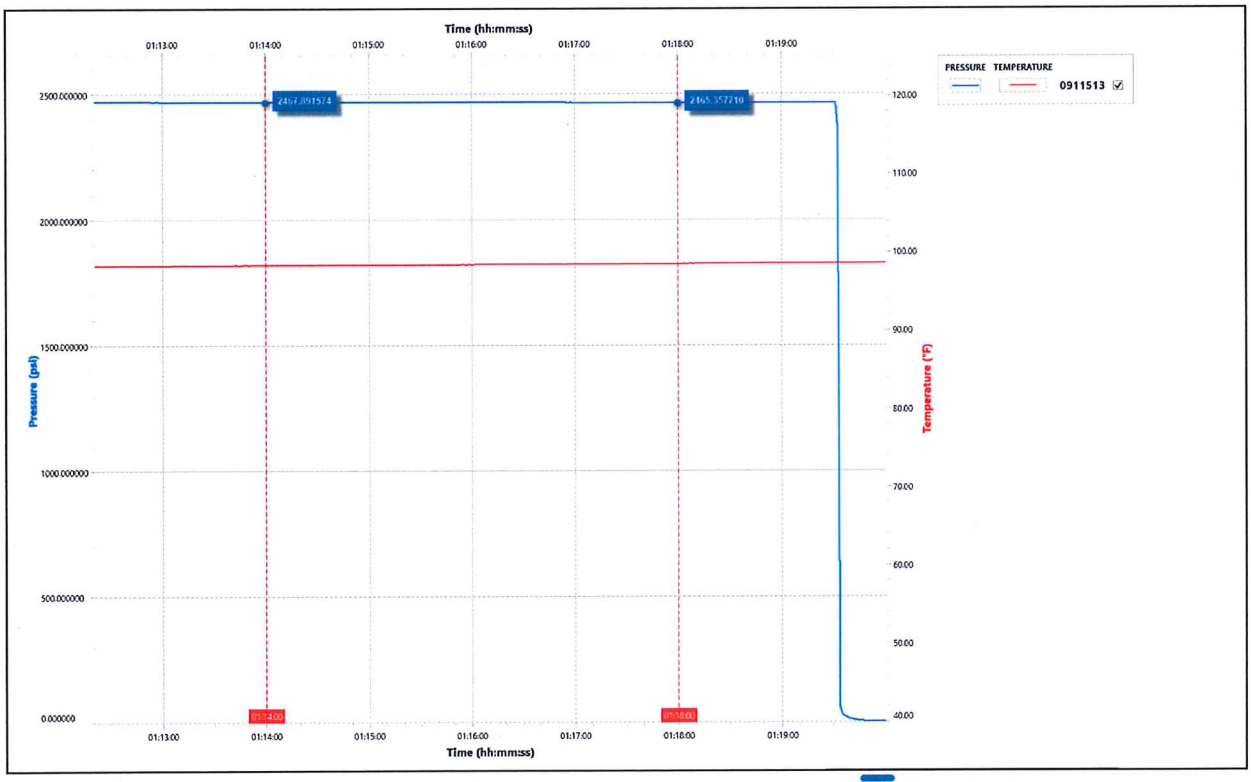
Customer:	**AIV, LP**		
Customer PO:	1177545	Serial Number:	61 - B90388
Work Order:	209127-01	Valve ID:	2HA947

Certification Information

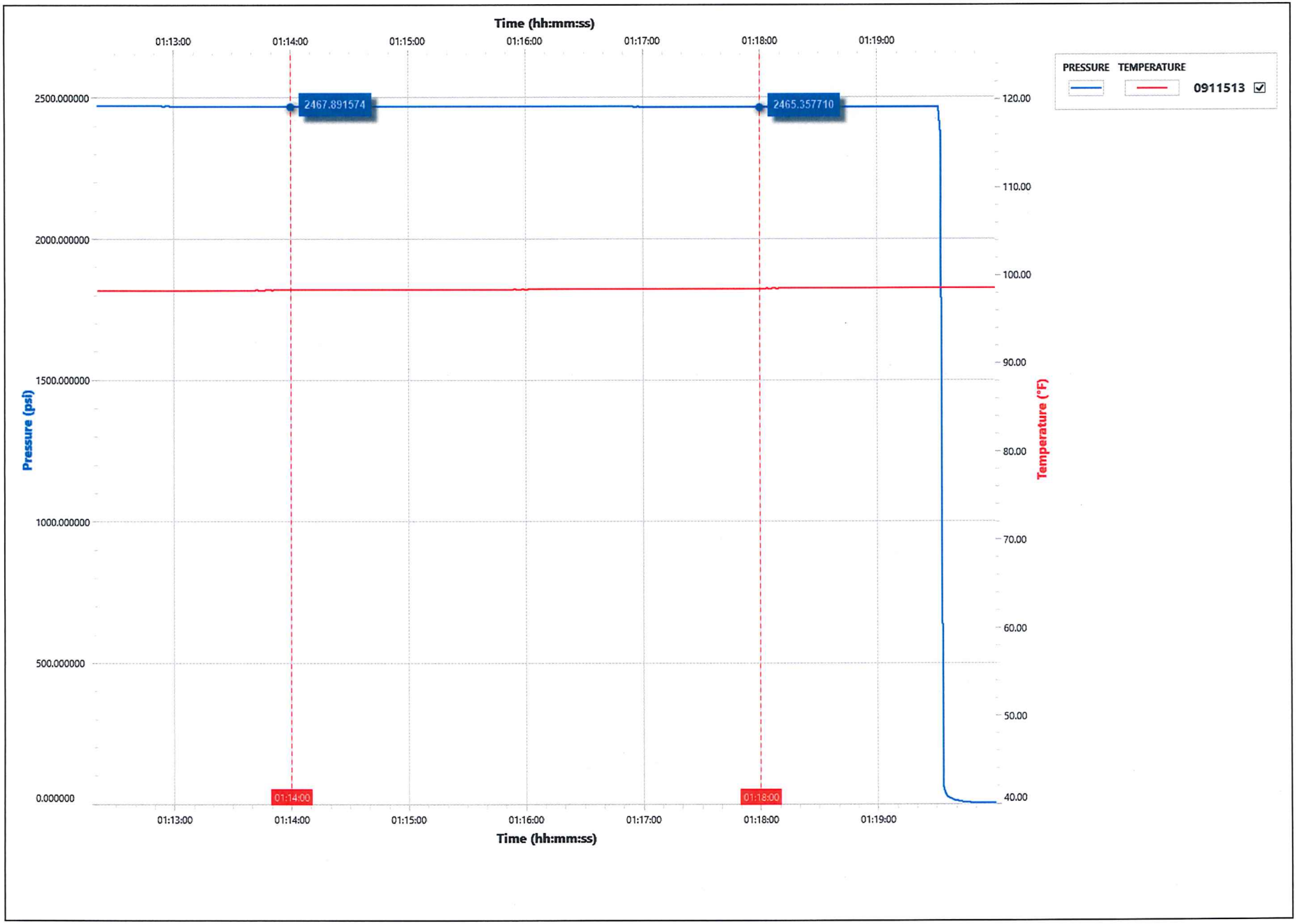
Cal. Device Used:	GCM1-0911513	Manufacturer:	ESI Technology Ltd.
Serial Number:	0911513	Calibration Date:	24/06/2026 10:45:12

Test Information

Test Date:	14/09/2026 15:50:19	Test Technician:	Ever Zavala
Procedure:	API 6D - HP Seat B	Test Medium:	Water
Test Pressure:	2,445 PSIG	Test Duration:	5 Min



Test Technician:	Ever Zavala	Signature:	Josh Alcoser Gulf Coast Modification Josh Alcoser Inspection Coordinator
Witness/Inspector:	Roger Johnson	Signature:	Roger Johnson Roger Johnson (Sep 23, 2026 13:57:08 CDT)


Time (hh:mm:ss)
01:13:00
01:14:00
01:15:00
01:16:00
01:17:00
01:18:00
01:19:00
PRESSURE
TEMPERATURE
0911513
2467.891574
2465.357710
2500.000000
2000.000000
1500.000000
1000.000000
500.000000
0.000000
Pressure (psi)
120.00
110.00
100.00
90.00
80.00
70.00
60.00
50.00
40.00
Temperature (°F)
01:14:00
01:18:00
Time (hh:mm:ss)

SPECIALLOY
PRESSURE TEMPERATURE TORQUE FLOW CALIBRATION

ISO 17025 CL-298

Specialloy Instrumentation
11702 Brittmoore Park
Houston, TX 77041
713-827-7776

Certificate of Calibration

SDCR

Certificate No. 062426-4 **Print Date** 6/24/2026

Unit Under Test (UUT)

UUT ID :	911513	Calibration Date:	6/24/2026
Model:	GS4200-USB1000AM-TEMP	Expiration Date:	12/24/2026
Range:	(0~15000)psi	Manufacturer:	ESI
Resolution:	1psi	Customer:	Gulf Coast Modification
Application Site:	Specialloy Calibration Laboratory	Address:	22806 NORTHWEST LAKE DR. HOUSTON, TX. 77095
Accuracy:	0.15% FS		

Calibration Results

	Result	Max Error	Temperature	Humidity	Atmospheric pressure
As Found	In tolerance	-0.09 %	70.2°F	57%	100.2 kPa
As Left	In tolerance	-0.09 %	70.2°F	57%	100.2 kPa

Calibration Data - As Found

Max Error: -0.09 %

	SetPoints psi	Reference psi	Gauge psi	Tolerance psi	Deviation psi	Error (%)	Result
	0.0	0.0	0.0	22.5	0.0	0.00	Pass
	3750.0	3752.0	3750.0	22.5	-2.0	-0.01	Pass
	7500.0	7503.0	7500.0	22.5	-3.0	-0.02	Pass
	11250.0	11257.0	11250.0	22.5	-7.0	-0.05	Pass
	15000.0	15013.0	15000.0	22.5	-13.0	-0.09	Pass
	11250.0	11257.0	11250.0	22.5	-7.0	-0.05	Pass
	7500.0	7506.0	7500.0	22.5	-6.0	-0.04	Pass
	3750.0	3754.0	3750.0	22.5	-4.0	-0.03	Pass
	0.0	0.0	0.0	22.5	0.0	0.00	Pass
TEMP	72.0	70.0	69.5	3.0	-0.5	0.00	Pass

=== As Found Calibration Data End ===

Calibration Data - As Left

Max Error: -0.09 %

	SetPoints psi	Reference psi	Gauge psi	Tolerance psi	Deviation psi	Error (%)	Result
	0.0	0.0	0.0	22.5	0.0	0.00	Pass
	3750.0	3752.0	3750.0	22.5	-2.0	-0.01	Pass
	7500.0	7503.0	7500.0	22.5	-3.0	-0.02	Pass
	11250.0	11257.0	11250.0	22.5	-7.0	-0.05	Pass
	15000.0	15013.0	15000.0	22.5	-13.0	-0.09	Pass
	11250.0	11257.0	11250.0	22.5	-7.0	-0.05	Pass
	7500.0	7506.0	7500.0	22.5	-6.0	-0.04	Pass
	3750.0	3754.0	3750.0	22.5	-4.0	-0.03	Pass
	0.0	0.0	0.0	22.5	0.0	0.00	Pass
TEMP	72.0	70.0	69.5	3.0	-0.5	0.00	Pass

=== As Left Calibration Data End ===

References

Manufacturer	Model	Serial Number	Certificate No.	Due Date	Accuracy
Additel Corporation	ADT681-05-GP15K-PSI-N	211H22620027	051326-3	5/13/2027	0.05%
---	---	---	---	---	---
---	---	---	---	---	---
TRACEABLE	6530	260988679	6530-16611970	3/26/2028	1%

Calibrated By:

Comments:

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The decision rule used is the Simple Acceptance as defined in ILAC G8 with measurement uncertainty value that will not exceed 10% of the tolerance.
Uncertainty of Measurement is stated with a coverage factor of k=2 with a confidence level of 95%. Specialloy ensures the measurements reported herein are traceable to the International System of Units (SI) through the National Institute of Standards and Technology (NIST). This traceability is established through a documented chain of calibrations with known uncertainties, ensuring the reliability of the reported values

Specialloy Instrumentation
11702 Brittmoore Park
Houston, TX 77041
713-827-7776

Certificate of Calibration

Certificate No. 042126-21 **Print Date** 4/21/2026

Unit Under Test (UUT)

UUT ID :	GCM1/10018	Calibration Date:	4/21/2026
Model:	40S-10K	Expiration Date:	10/21/2026
Range:	(0~10000)psi	Manufacturer:	Stewarts USA
Resolution:	100psi	Customer:	Gulf Coast Modification LP
Application Site:	Specialloy Calibration Laboratory	Address:	22806 NORTHWEST LAKE DR.HOUSTON, TEXAS 77095
Accuracy:	0.5% FS		

Calibration Results

	Result	Max Error	Temperature	Humidity	Atmospheric pressure
As Found	In tolerance	-0.44 %	70°F	50%	1.016 kPa
As Left	In tolerance	-0.44 %	70°F	50%	1.016 kPa

Calibration Data - As Found

Max Error: -0.44 %

SetPoints psi	Reference psi	Gauge psi	Tolerance psi	Deviation psi	Error (%)	Result
0.0	0.0	0.0	50.0	0.0	0.00	Pass
2500.0	2500.0	2500.0	50.0	0.0	0.00	Pass
5000.0	5036.0	5000.0	50.0	-36.0	-0.36	Pass
7500.0	7543.0	7500.0	50.0	-43.0	-0.43	Pass
10000.0	10044.0	10000.0	50.0	-44.0	-0.44	Pass
7500.0	7476.0	7500.0	50.0	24.0	0.24	Pass
5000.0	4968.0	5000.0	50.0	32.0	0.32	Pass
2500.0	2463.0	2500.0	50.0	37.0	0.37	Pass
0.0	0.0	0.0	50.0	0.0	0.00	Pass

=== As Found Calibration Data End ===

Calibration Data - As Left

Max Error: -0.44 %

SetPoints psi	Reference psi	Gauge psi	Tolerance psi	Deviation psi	Error (%)	Result
0.0	0.0	0.0	50.0	0.0	0.00	Pass
2500.0	2500.0	2500.0	50.0	0.0	0.00	Pass
5000.0	5036.0	5000.0	50.0	-36.0	-0.36	Pass
7500.0	7543.0	7500.0	50.0	-43.0	-0.43	Pass
10000.0	10044.0	10000.0	50.0	-44.0	-0.44	Pass
7500.0	7476.0	7500.0	50.0	24.0	0.24	Pass
5000.0	4968.0	5000.0	50.0	32.0	0.32	Pass
2500.0	2463.0	2500.0	50.0	37.0	0.37	Pass
0.0	0.0	0.0	50.0	0.0	0.00	Pass

=== As Left Calibration Data End ===

References

Manufacturer	Model	Serial Number	Certificate No.	Due Date	Accuracy
Additel Corporation	ADT681-05-GP15K-PSI-N	211H22620027	N25052240	5/21/2026	0.05%
---	---	---	---	---	---
---	---	---	---	---	---
Acu-rite	01121M	204651	1359293	5/13/2026	1%

Comments:

Calibrated By:

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Specialloy Instrumentation
11702 Brittmoore Park
Houston, TX 77041
713-827-7776

Certificate of Calibration

Certificate No. 051126-7 **Print Date** 5/11/2026

Unit Under Test (UUT)

UUT ID :	GCM1/6021
Model:	40S-6K
Range:	(0~6000)psi
Resolution:	100psi
Application Site:	Specialloy Calibration Laboratory
Accuracy:	0.5% FS

Calibration Date:	5/11/2026
Expiration Date:	11/11/2026
Manufacturer:	Stewarts USA
Customer:	Gulf Coast Modification LP
Address:	22806 NORTHWEST LAKE DR. HOUSTON, TEXAS 77095

Calibration Results

	Result	Max Error	Temperature	Humidity	Atmospheric pressure
As Found	In tolerance	-0.26 %	69°F	64%	1.006 kPa
As Left	In tolerance	-0.26 %	69°F	64%	1.006 kPa

Calibration Data - As Found

Max Error: -0.26 %

SetPoints psi	Reference psi	Gauge psi	Tolerance psi	Deviation psi	Error (%)	Result
0.0	0.0	0.0	30.0	0.0	0.00	Pass
1500.0	1505.0	1500.0	30.0	-5.0	-0.08	Pass
3000.0	3006.5	3000.0	30.0	-6.5	-0.11	Pass
4500.0	4508.7	4500.0	30.0	-8.7	-0.14	Pass
6000.0	6015.6	6000.0	30.0	-15.6	-0.26	Pass
4500.0	4508.2	4500.0	30.0	-8.2	-0.14	Pass
3000.0	2996.6	3000.0	30.0	3.4	0.06	Pass
1500.0	1496.4	1500.0	30.0	3.6	0.06	Pass
0.0	0.0	0.0	30.0	0.0	0.00	Pass

=== As Found Calibration Data End ===

Calibration Data - As Left

Max Error: -0.26 %

SetPoints psi	Reference psi	Gauge psi	Tolerance psi	Deviation psi	Error (%)	Result
0.0	0.0	0.0	30.0	0.0	0.00	Pass
1500.0	1505.0	1500.0	30.0	-5.0	-0.08	Pass
3000.0	3006.5	3000.0	30.0	-6.5	-0.11	Pass
4500.0	4508.7	4500.0	30.0	-8.7	-0.14	Pass
6000.0	6015.6	6000.0	30.0	-15.6	-0.26	Pass
4500.0	4508.2	4500.0	30.0	-8.2	-0.14	Pass
3000.0	2996.6	3000.0	30.0	3.4	0.06	Pass
1500.0	1496.4	1500.0	30.0	3.6	0.06	Pass
0.0	0.0	0.0	30.0	0.0	0.00	Pass

=== As Left Calibration Data End ===

References

Manufacturer	Model	Serial Number	Certificate No.	Due Date	Accuracy
Additel Corporation	ADT681-02-GP6K-PSI-N	211H230F0027	051425-11	5/14/2026	0.02%
---	---	---	---	---	---
---	---	---	---	---	---
Acu-rite	01121M	204651	1359293	5/13/2026	1%

Comments:

Calibrated By:

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Specialloy Instrumentation
11702 Brittmoore Park
Houston, TX 77041
713-827-7776

ISO 17025 CL-286

Certificate of Calibration
TORQUE WRENCH

Certificate No. ST2-081226-6 **Print Date** 8/12/2026

Unit Under Test (UUT)

UUT ID :	TWD-GCM1-21	Calibration Date:	8/12/2026
Model:	100 FT/LBS DUAL	Expiration Date:	2/12/2027
Range:	(0~100)FT/LBS	Manufacturer:	CDI
Resolution:	2FT/LBS	Customer:	Gulf Coast Modification
Application Site:	Specialloy Calibration Laboratory	Address:	22806 NORTHWEST LAKE DR.HOUSTON, TEXAS 77095
Accuracy:	4% FS		

Calibration Results

	Result	Max Error	Temperature	Humidity	Atmospheric pressure
As Found	In tolerance	4.00 %	69°F	43%	1.014 kPa
As Left	In tolerance	4.00 %	69°F	43%	1.014 kPa

Calibration Data - As Found

Max Error: 4.00 %

	SetPoints FT/LBS	Reference FT/LBS	Gauge FT/LBS	Tolerance FT/LBS	Deviation FT/LBS	Error (%)	Result
CW	25.0	25.2	25.0	4.0	-0.2	-0.20	Pass
	25.0	25.5	25.0	4.0	-0.5	-0.50	Pass
	25.0	25.5	25.0	4.0	-0.5	-0.50	Pass
	60.0	60.4	60.0	4.0	-0.4	-0.40	Pass
	60.0	60.6	60.0	4.0	-0.6	-0.60	Pass
	60.0	60.6	60.0	4.0	-0.6	-0.60	Pass
	100.0	103.6	100.0	4.0	-3.6	-3.60	Pass
	100.0	103.8	100.0	4.0	-3.8	-3.80	Pass
	100.0	103.5	100.0	4.0	-3.5	-3.50	Pass
CCW	25.0	25.4	25.0	4.0	-0.4	-0.40	Pass
	25.0	25.4	25.0	4.0	-0.4	-0.40	Pass
	25.0	25.2	25.0	4.0	-0.2	-0.20	Pass
	60.0	58.1	60.0	4.0	1.9	1.90	Pass
	60.0	58.1	60.0	4.0	1.9	1.90	Pass
	60.0	58.2	60.0	4.0	1.8	1.80	Pass
	100.0	96.0	100.0	4.0	4.0	4.00	Pass
	100.0	96.2	100.0	4.0	3.8	3.80	Pass
	100.0	96.2	100.0	4.0	3.8	3.80	Pass

=== As Found Calibration Data End ===

Calibration Data - As Left

Max Error: 4.00 %

	SetPoints FT/LBS	Reference FT/LBS	Gauge FT/LBS	Tolerance FT/LBS	Deviation FT/LBS	Error (%)	Result
CW	25.0	25.2	25.0	4.0	-0.2	-0.20	Pass
	25.0	25.5	25.0	4.0	-0.5	-0.50	Pass
	25.0	25.5	25.0	4.0	-0.5	-0.50	Pass
	60.0	60.4	60.0	4.0	-0.4	-0.40	Pass
	60.0	60.6	60.0	4.0	-0.6	-0.60	Pass
	60.0	60.6	60.0	4.0	-0.6	-0.60	Pass
	100.0	103.6	100.0	4.0	-3.6	-3.60	Pass
	100.0	103.8	100.0	4.0	-3.8	-3.80	Pass
	100.0	103.5	100.0	4.0	-3.5	-3.50	Pass
CCW	25.0	25.4	25.0	4.0	-0.4	-0.40	Pass
	25.0	25.4	25.0	4.0	-0.4	-0.40	Pass
	25.0	25.2	25.0	4.0	-0.2	-0.20	Pass
	60.0	58.1	60.0	4.0	1.9	1.90	Pass
	60.0	58.1	60.0	4.0	1.9	1.90	Pass
	60.0	58.2	60.0	4.0	1.8	1.80	Pass
	100.0	96.0	100.0	4.0	4.0	4.00	Pass
	100.0	96.2	100.0	4.0	3.8	3.80	Pass
	100.0	96.2	100.0	4.0	3.8	3.80	Pass

=== As Left Calibration Data End ===

References

Manufacturer	Model	Serial Number	Certificate No.	Due Date	Accuracy
CDI	2503-F-DTT 25-250 FT/LBS	1117880068	T 45510.57163	8/16/2027	0.5%
---	---	---	---	---	---
---	---	---	---	---	---
TRACEABLE	6530	260988679	6530-16611970	3/26/2028	3%

michael Stewart
Calibrated By:

Comments:

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